

Shiye Du

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EDUCATION

Westlake University

Bachelor of Science, Chemistry

Hangzhou, China

July 2023 - July 2027 (Expected)

- **GPA:** 4.070/4.300, 1st in Chemistry cohort

Duke University

Visiting International Student, Department of Chemistry

Durham, NC, USA

Aug. 2025 - Dec. 2025

- **Relevant coursework:** Elementary Differential Equations; Discrete Mathematics for Computer Science; Introduction to Data Science.

RESEARCH EXPERIENCE

Professor Hongyu Chen's Lab, Westlake University

Undergraduate Researcher

Hangzhou, China

Jan. 2025 - Present

Co-supervised by Prof. Mouhong Lin, College of Optical Science and Engineering, Zhejiang University.

Project: Polarization-Polar-Vector Force Law for Optically Axled Nanorotors

- Developing a first-author manuscript that establishes a polarization-polar-vector force law for optically axled nanorotors.
- Built high-throughput FDTD/MST workflows to compute internal rotor forces in ring, shell, and eccentric-stator geometries.
- Demonstrated CPL-driven nonconservative orbital work and LP-driven conservative angular locking in achiral spherical rotors.
- Combined material screening and Bayesian optimization to select Ag shell rotors, reducing simulated $\Delta T_{\max}$ from ~ 65 to ~ 15 K at 10^9 W m⁻² while maintaining sufficient orbital driving force.

Project: DNA-Clutched Photothermally Reconfigurable Multishell Plasmonic Nanomachines

- Co-leading the computational design and modeling component of a collaborative manuscript in preparation on DNA-clutched Au-SiO₂-Au multishell plasmonic nanomachines.
- Developed FDTD and descriptor-assisted workflows to design photothermally switchable concentric/eccentric multishell states driven by DNA dehybridization-mediated shell displacement.
- Predicted state-dependent plasmonic coupling, hot-spot redistribution, absorption heating, and Raman-relevant near-field signatures to guide ongoing experimental synthesis by PhD collaborators.

Project: Electromagnetic Descriptor-Mediated Inverse Design of Colloidal Plasmonic Nanostructures

- Co-leading a manuscript in preparation on electromagnetic descriptors that serve as a physics-informed intermediate layer for prediction and inverse design of colloidal plasmonic nanostructures.
- Developed a structure-descriptor-spectrum learning framework, enabling a shared descriptor-to-spectrum decoder across nanoparticle geometries while retraining only the structure-to-descriptor encoder for new structures.
- Improved cross-geometry prediction and interpretability over direct structure-spectrum mapping by using descriptors such as hot-spot volume, normal-field volume, multipolar amplitudes, and absorption-related field integrals.

Professor Jianjun Cheng's Lab, Westlake University
Undergraduate Researcher

Hangzhou, China
Sept. 2023 - Jan. 2025

Project: Rapid NCA Ring-Opening Polymerization of Polypeptides

- Synthesized BLG-NCA monomers and PBLG polypeptides, and monitored polymerization kinetics using in situ FTIR spectroscopy.
- Investigated macrodipole-assisted macroinitiators as a strategy to accelerate NCA ring-opening polymerization.
- Gained training in monomer synthesis, controlled polymerization, reaction monitoring, and polymer characterization.

International Genetically Engineered Machine (iGEM) Competition

Team Member

Hangzhou, China
Dec. 2023 - Nov. 2024

- Designed components of an E. coli genetic circuit for tumor-cell recognition and inducible therapeutic-response activation.
- Contributed to receptor-engineering and project documentation for a synthetic-biology project awarded an iGEM Silver Medal.

AWARDS AND HONORS

- National Scholarship, Ministry of Education of China, 2025
- Zhejiang Provincial Government Scholarship, 2024
- Outstanding Student Award, Westlake University, 2025
- Learning Innovation Award, Westlake University, 2024 & 2025
- Outstanding Contribution Award, Student Ambassador Program, Westlake University, 2024
- Selected Campus Arts Honors: Westlake University Top Ten Singer Finalist; Best Singer and Best Performance Awards, 2024

LEADERSHIP & SERVICE

Student Development Center, Westlake University

Head, Academic Department

Hangzhou, China
2024 - Present

- Led academic-support initiatives and peer-learning activities for undergraduate students.

Westlake University Student Ambassador Program

Student Ambassador

Hangzhou, China
2023 - 2024

- Represented Westlake University in admissions outreach, campus visits, and prospective-student communication.

SKILLS

Computational Nanophotonics: Lumerical FDTD; Maxwell-stress-tensor optical force/torque calculations; high-throughput electromagnetic simulation; photothermal modeling; multipolar and near-field analysis

Machine Learning & Data Analysis: Python; PyTorch; Gaussian-process Bayesian optimization; SHAP interpretability analysis; automated simulation-data extraction

Experimental & Characterization Skills: polypeptide synthesis; basic organic synthesis; colloidal nanoparticle synthesis; TEM characterization; in situ FTIR kinetics

Languages: Mandarin Chinese, native; English, fluent (IELTS 7.0)